

3. (a) (i) Air bags help to keep passengers safe in the event of an accident.

Underline the words that correctly complete the sentences below.

[3]

Air bags (**increase** / **decrease** / **remove**) the time it takes to reduce the passengers' (**momentum** / **inertia** / **mass**) to zero.

This (**increases** / **decreases** / **doubles**) the force acting on them.

- (ii) Air bags are one safety feature of cars designed to reduce harm to passengers. Name **one** other safety feature.

[1]

.....

- (b) Explain why cars are designed to have a more aerodynamic shape.

[2]

.....

.....

.....

